

Botox Markdown Suite

Native LaTeX-Grade Typesetting & Slide Decks in a Single Binary

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Abstract – *Botox is a pure Rust, single-binary command-line suite that compiles standard Markdown documents into publication-grade LaTeX-style PDF documentation or presentation slide decks. Featuring an embedded in-process typesetting engine, authentic New Computer Modern typography, advanced TeX mathematics, native IEEE-standard bibliographies, and Pandoc extensions, Botox requires zero external runtime dependencies.*

Keywords – *Document Engineering, LaTeX Typesetting, Pandoc Compatibility, Rust*

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1 Overview and Architecture

Botox provides a single, self-contained native executable (`botox`) that transforms standard Markdown files into either publication-grade LaTeX-style PDF documents or clean 16:9 presentation slide decks.

Mode detection is automatic based on frontmatter structure (or can be explicitly declared with `--pdf` or `--slides`). This `README.md` file itself serves as a comprehensive living specification of all supported features and can be compiled directly:

```
botox README.md -o README.pdf
```

1.1 Architectural Pillars

Unlike traditional document compilation pipelines that require a web of external runtimes:

- **Zero Runtime Dependencies:** No Node.js, no Python, no Pandoc runtime, no headless Chrome, and no multi-gigabyte TeX Live distributions.
- **Embedded Typesetting Engine:** Built on the [Rust Language \[1\]](#) with the [Typst \[2\]](#) layout engine and New Computer Modern font family (Roman, Bold, Italic, and Math) compiled directly into the binary¹.
- **Instantaneous Compilation:** Sub-second compilation times (typically 400ms to 900ms) for multi-page documents.
- **Cross-Platform Portability:** Builds as a single static executable for Linux, macOS, and Windows.

1.2 Feature Matrix

- [x] Standalone native single binary (`botox`) with zero external requirements
- [x] Complete Pandoc frontmatter compatibility (`geometry` , `lang` , `toc-depth` , `number-sections`)
- [x] Typography extensions: subscripts (`H~2~0`), superscripts (`10^6^`), strikethrough (`~~text~~`)
- [x] Native multi-pass footnotes with bottom-of-page rule separation
- [x] Definition lists and GFM task checklists
- [x] Advanced TeX mathematical notation (matrices, piecewise systems, partial derivatives, accents)
- [x] Pipe tables with column alignments (`:---` , `:---:` , `---:`) and cell formatting
- [x] Automatic IEEE-standard numbered bibliography generated from hyperlinks
- [x] High-resolution figures and vector graphics (PNG, JPEG, SVG, WebP, GIF) with automated captions and numbering
- [x] Marp-compatible 16:9 presentation slide decks partitioned by `---`

¹Embedded font assets include 17 New Computer Modern and DejaVu typeface variants loaded directly from in-memory binary slices.

- [x] Cascading configuration hierarchy (`botox.yaml`)

2 Installation and Quick Start

2.1 System Binary Installation

The pre-compiled binary can be copied directly to any folder on your system `PATH` :

```
# Install to user local binary path
cp bin/botox ~/.local/bin/botox
```

2.2 Building from Source

To compile the standalone binary from source using the Rust toolchain:

```
cd botox
cargo build --release
strip target/release/botox
cp target/release/botox ~/.local/bin/botox
```

2.3 Unified CLI Invocation

The single executable `botox` handles both documents and presentations:

```
# Initialize a new document or presentation with active default settings
botox init document.md
botox init slides.md

# Compile documentation PDF (auto-detected)
botox README.md -o README.pdf

# Compile presentation slides (auto-detected)
botox slides.md -o slides.pdf

# Transform hyperlinks into an IEEE Bibliography
botox document.md --bib -o document.pdf

# Toggle automatic Table of Contents
botox document.md --toc -o document.pdf
botox document.md --no-toc -o document.pdf

# Inspect active configuration settings
botox config
```

3 Typography and Pandoc Extensions

Botox adheres to standard Pandoc Markdown conventions and extensions.

3.1 Inline Styling and Formatting

Standard emphasis, bold, code, and strikethrough are fully supported alongside intra-word subscript and superscript delimiters:

- **Chemical Formulas & Exponents:** Water is written as H_2O , glucose as $\text{C}_6\text{H}_{12}\text{O}_6$, and astronomical orders of magnitude as 10^6 or 10^{-19} .
- **Strikethrough:** Deprecated workflows are marked with ~~legacy pipelines~~ and replaced with instant native compilation.
- **Code Spans:** Variables and registers like `reg_ex_mem` and `instruction_pointer` are rendered with monospace background badges.

3.2 Definition Lists

Pandoc-style definition lists format technical glossaries with bold headwords and indented definitions:

Zero-Dependency Architecture A self-contained executable that bundles all fonts, engines, and converters without external interpreters.

Forwarding Unit A microarchitectural hazard-mitigation bypass that routes execution results directly between pipeline stages.

Automatic Bibliography In-text citations and end-of-document reference sections generated automatically from standard Markdown links.

3.3 Blockquotes

Academic citations and callouts render with subtle left border rules and margins:

““Simplicity is prerequisite for reliability.” — Edsger W. Dijkstra, *Selected Writings on Computing*”

3.4 Callout Divs

Pandoc-style fenced divs (`:::`) render as color-coded advisory blocks with distinct visual borders:

Note

Pandoc `link_attributes`: The `{width=50%}` syntax originates from Pandoc’s `link_attributes` extension. Botox natively parses and renders image attributes without requiring any external preprocessors.

Tip

Use cross-references like `@fig:arch`, `@tbl:perf`, `@sec:math`, or `@eq:energy` to generate clickable hyperlinks that automatically track section and asset numbering.

Warning

Never mix unnumbered headings with strict numerical cross-references. Unnumbered headings (`{-}` or `{.unnumbered}`) are excluded from the numbering counter.

3.5 Figures, Graphics & Attributes

Raster and vector image assets (PNG, JPEG, SVG, WebP, GIF) are rendered natively without external decoders. Standard Markdown syntax with alt text generates numbered LaTeX figures with captions:

```
![Botox Architecture Pipeline](figures/architecture.png){width=65% #fig:arch}  
{width=10cm height=5cm}  
![Relative Scaling](banner.png){width=0.8\linewidth}
```

Attributes inside `{...}` support:

- Dimensions: `%` (e.g. `width=50%`), absolute units (`10cm`, `4in`, `200pt`, `100mm`), pixels (`300px`), and TeX factors (`0.8\linewidth`, `\textwidth`).
- Identifiers: `#fig:id` or `id=fig:id` for cross-referencing with `@fig:id`.

3.6 Cross-Referencing

Botox implements Pandoc cross-referencing for figures, tables, sections, and equations:

- `@fig:arch` references Figure 1
- `@tbl:perf` references Table 1 (see Table 1)
- `@sec:math` references Section 4 (see Section 4)
- `@eq:energy` references Equation 1 (see Equation 1)

3.7 Page Breaks

LaTeX and Pandoc page break commands placed anywhere in the document are translated into native page divisions:

- `\newpage`
- `\pagebreak`
- `\clearpage`
- `<!-- pagebreak -->`

4 Advanced Mathematical Typography

Mathematical expressions render with full LaTeX fidelity using embedded New Computer Modern Math fonts.

4.1 Fractions, Accents, and Operators

Nested braces in fractions and roots resolve recursively with proper font scaling:

$$E = mc^2 + \frac{1}{2}mv^2 \tag{1}$$

$$\vec{F} \approx m \frac{\partial^2 \vec{r}}{\partial t^2} + \sqrt[3]{\frac{\alpha_{11}}{\beta + \gamma}} \pm \vec{\varepsilon} \tag{2}$$

Definite integrals, infinite series, and limits preserve standard limits and display styles:

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}, \quad \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x, \quad \lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1 \quad (3)$$

4.2 Matrix Environments

Parenthesized (`pmatrix`), bracketed (`bmatrix`), and determinant (`vmatrix`) matrices are fully supported:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \left| \begin{array}{cc} \lambda - a & b \\ c & \lambda - d \end{array} \right| \quad (4)$$

4.3 Piecewise Cases

Piecewise function definitions with conditions and text descriptions render using the `cases` environment:

$$f(x) = \begin{cases} \sqrt{x} & \text{if } x \geq 0 \\ -\sqrt{-x} & \text{if } x < 0 \end{cases} \quad (5)$$

5 Structured Tables

Pipe tables support header alignment markers (`:---` for left, `:---:` for center, and `---:` for right), automated captions, and cross-reference labels:

Tool	Pipeline Stage	Target Output	Performance
Botox Native Engine	Typst Layout	PDF Vector	< 1.0 s
Rust Compiler [1]	Codegen	Native ELF / PE	$O(N)$
LLVM Backend [3]	Optimization	Machine Code	Parallel
Typst Engine [2]	Typesetting	PDF / SVG	Real-time

Table 1: Pipeline execution comparison across toolchains.

Cells seamlessly support inline code, formatting, mathematics, and web links.

6 Automatic Bibliography Transformation

When `bibliography: true` is set in the frontmatter (or enabled via the `-b / --bibliography` command-line flag), all standard web links are transformed into an **IEEE-standard numbered bibliography**:

- **In-Text Citations:** Hyperlinks in the body receive bracketed citations (e.g. `[1]`, `[2]`). Clicking the citation number jumps directly to the corresponding entry in the bibliography.
- **Deduplication:** Repeated links to the same destination automatically share the same citation number across the entire document.

- **References Section:** An unnumbered References section is generated at the end of the document featuring hanging indents and electronic resource citations (e.g. including the [GitHub Platform \[4\]](#)).
- **Localization:** Respects the document `lang` parameter (e.g., *References*, *Références*, or *Literaturverzeichnis*).

7 Presentation Slide Decks

Slide decks are authored in the same Markdown files, using horizontal rules (`---`) as slide boundaries. When `marp: true` or slide themes are detected in the frontmatter, Botox compiles the document into a 16:9 presentation PDF:

```
---
marp: true
theme: default
paginate: true
backgroundColor: "#f8fafc"
color: "#0f172a"
---

# Microarchitecture Hazards
Eliott JAQUIER, Shanshe GUNDISHVILI

---

# Forwarding Unit Implementation
- Direct bypass from EX/MEM to ID/EX
- Zero stall cycles for ALU dependencies
- Fallback stall insertion for Load-Use hazards
```

Built-in slide themes include `default`, `academic`, `dark`, and `nord`, with optional custom `backgroundColor` and `color` hex overrides.

8 Configuration Reference

Botox loads configuration settings from a cascading hierarchy:

- CLI argument `--config <path>`
- Project-level configuration file: `./botox.yaml`
- User-level global configuration file: `~/.config/botox/config.yaml`

```
document:
  author: "Minus"
  affiliation: "Systems Engineering"
  fontsize: "11pt"
  mainfont: "New Computer Modern"
  mathfont: "New Computer Modern Math"
  papersize: "a4"
  columns: 1
  margin:
```

```

    x: "2.5cm"
    y: "2.5cm"
    section_numbering: true
    toc: true
    bibliography: true

slides:
  theme: "default"
  paginate: true

```

8.1 CLI Option Summary

Flag	Parameter	Description
<code>-o, --output</code>	<code><file></code>	Output PDF path (default: <code><input>.pdf</code>)
<code>-b, --bib</code>	None	Enable automatic IEEE Bibliography generation
<code>--no-bib</code>	None	Disable automatic Bibliography generation
<code>--toc</code>	None	Force generation of Table of Contents
<code>--no-toc</code>	None	Disable Table of Contents
<code>--author</code>	<code><name></code>	Override author name
<code>--font</code>	<code><name></code>	Override primary typeface
<code>-N, --number-sections</code>	None	Enable numbered section headings
<code>--pdf</code>	None	Force PDF documentation mode
<code>--slides</code>	None	Force presentation slide deck mode
<code>--config</code>	<code><file></code>	Supply custom configuration YAML path
<code>init</code>	<code><filename></code>	Initialize new document or slide deck with default settings
<code>config</code>	Subcommand	Display currently active configuration

Appendix

This appendix section demonstrates an unnumbered heading created with Pandoc’s `{-}` attribute syntax. It appears in the document structure without incrementing the section counter.

References

- [1] “Rust Language”, [Online]. Available: <https://www.rust-lang.org>.
- [2] “Typst”, [Online]. Available: <https://typst.app>.

- [3] “LLVM Backend”, [Online]. Available: <https://llvm.org>.
- [4] “GitHub Platform”, [Online]. Available: <https://github.com>.